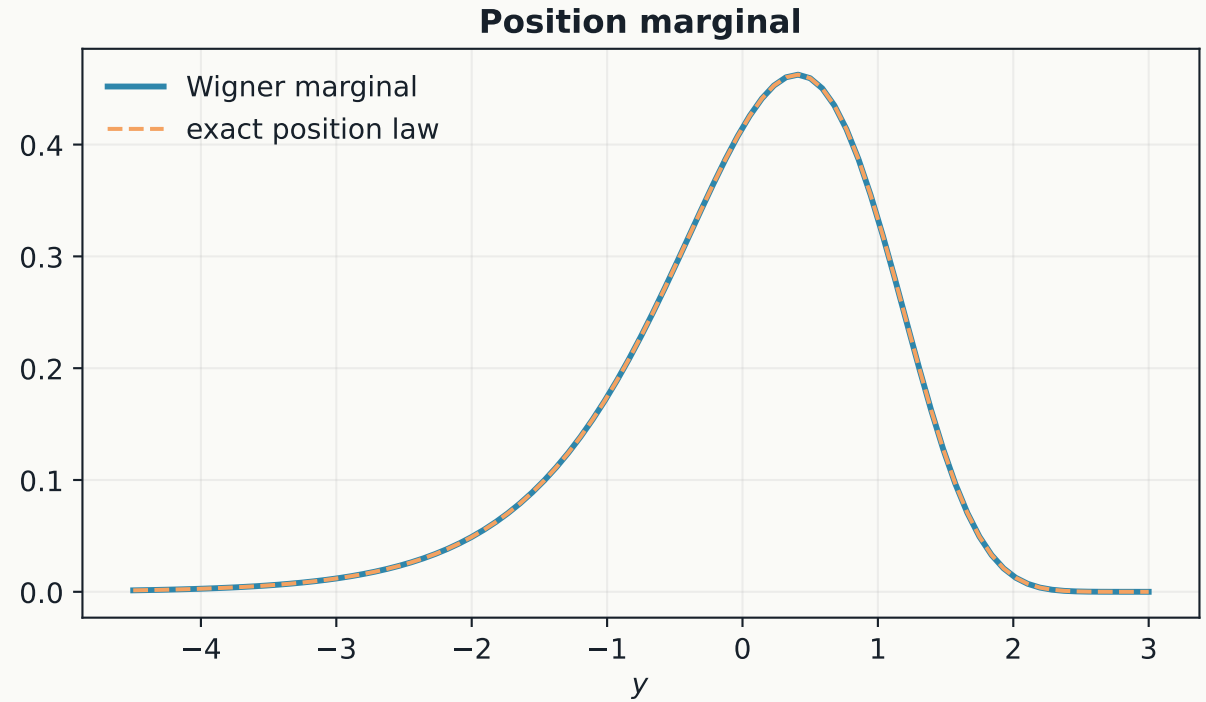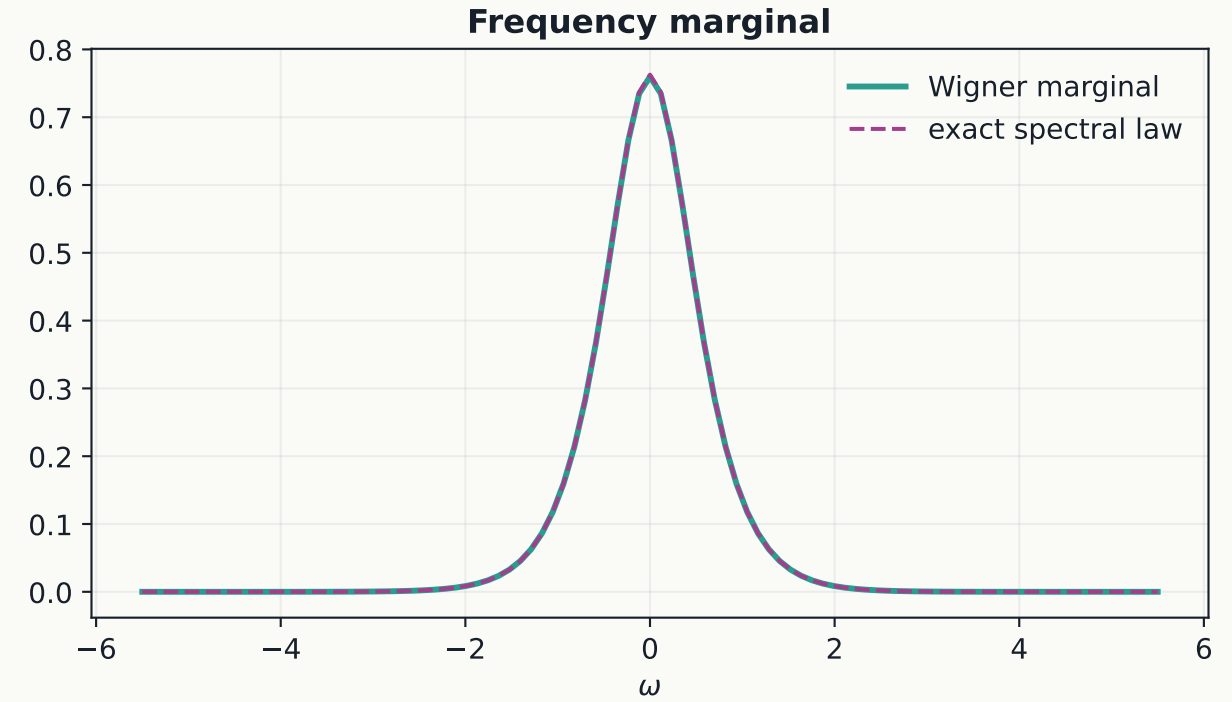The Wigner packet recovers both exact probability marginals